High-ROIC Compounding under AAOIFI Debt Limits: An Exploratory Backtest of Halal ROIC Reinvestment, 2022–2024

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Paper 02 · Exploratory note

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Abstract

This is an internal exploratory note, not a finished proof and not a live-return target. From 31 October 2022 through 30 December 2024, a cap-weighted basket of AAOIFI-screened S&P 500 names with return on invested capital (ROIC) above 15% and high reinvestment rates returned **39.6%** compound annual growth, versus **22.9%** for the S&P 500 (SPY) and **27.4%** for a Halal large-cap ETF (SPUS). The live rule pairs a hard ROIC floor with the top half of that pool by reinvestment rate—the "reinvestment engine"—and weights by market cap. Inside the high-ROIC pool, the high-reinvestment half beat the low-reinvestment half (43.2% vs 28.7% CAGR), which supports the compounding hypothesis more than the ROIC quintiles alone (Q1 did not beat Q5 cleanly). The main book is still a Halal mega-cap tech/platform portfolio: technology plus communication services average about 82% of GICS weight. A robustness sleeve restricted to SaaS, Healthcare, and MedTech returned 27.6% CAGR—roughly in line with SPUS, not a clear upgrade over the broad compounder book. Yahoo annual statements and a short live window limit how much we can claim about long-run persistence in capital-light sectors.

Contents

Status. Exploratory. Reproduce from `code.ipynb`. Figures in `figures/`. Sharia screens are a hard point-in-time constraint, not an overlay.

1. Hypothesis & Theory

1.1 What we are testing

The lab backlog item for this paper is return on invested capital (ROIC) paired with reinvestment rates among low-debt Halal equities [2]. The economic claim is that AAOIFI leverage limits already remove highly indebted firms; among what remains, businesses that **earn a high return on capital and plough a large share of those earnings back into growth** should compound faster than the broad market—especially in capital-light sectors like SaaS, Healthcare, and MedTech where R&D and intangible reinvestment matter more than plant and equipment.

> After dropping banned businesses and names above the AAOIFI debt cap, do Halal companies with ROIC $> 15\%$ and high reinvestment rates hold up as a compounding book when owned in proportion to size?

ROIC measures how hard each dollar of invested capital works: ROIC = NOPAT/Invested Capital, with NOPAT = EBIT $\times(1 - \text{tax rate})$. Reinvestment rate measures how much of that NOPAT is put back into the business:

$$\text{Reinvestment rate} = \frac{\text{CapEx} - \text{D\&A} + \Delta\text{NWC} + \text{R\&D}}{\text{NOPAT}}.$$

R&D is counted as reinvestment so capital-light software and med-tech names are not scored as "low reinvestment" simply because they spend little on factories.

1.2 Why ROIC and reinvestment might interact with Halal screens

AAOIFI financial screens cap interest-bearing debt, cash, and receivables relative to market value. In this test the debt rule is total debt below 30% of trailing 24-month market cap, plus the library's cash and receivables screens. The hypothesis is that those caps leave a cleaner pool where return *on* capital—not cheapness on capital—is more informative.

We do *not* claim that Halal screens create a ROIC premium by themselves. The live window is short, mega-cap-heavy, and partly overlaps the 2023–2024 AI/platform boom. The capital-light robustness sleeve is included to test whether SaaS / Healthcare / MedTech compounders persist; it is not the primary live rule.

1.3 Universe and screening standard

Item	Choice
Target factors	ROIC > 15%; reinvestment rate in top half of that pool
Starting universe	Current S&P 500 list (503 tickers)
After banned-business screen	439 tickers (64 removed)
Focus-sector tags (diagnostic)	SaaS 40, Healthcare 25, MedTech 33, Other 341
Screening standard	AAOIFI financial ratios, not DJIM
Halal benchmark	SPUS (SP Funds S&P 500 Sharia Industry Exclusions)
All-stock benchmark	SPY (S&P 500)
Sample requested	1 Jan 2019 – 31 Dec 2024
First metrics date	30 Nov 2021
Live window used for stats	31 Oct 2022 – 30 Dec 2024 (544 trading days)

Sector exclusions (banks, conventional insurance, alcohol, tobacco, gambling, weapons, and similar) run once on the current constituent list. Debt, cash, and receivables screens re-run at every month-end on point-in-time filings. ROIC and reinvestment come from yfinance annual statements with a 90-day publication lag.

SPUS is the Halal *index* comparison required by the lab template. It is an industry-exclusion ETF, cap-weighted, not identical to a pure AAOIFI financial-ratio index.

Yahoo had no price history in-sample for Q, SNDK, and FDXF; BKR hit a cache lock once during download. They are skipped where data are missing.

1.4 What would count as a finished proof

This note would move from "exploratory" to "ready for a live mandate" only after: a frozen rule, ten to fifteen years of SEC income/cash-flow tags, point-in-time S&P membership, a down year fully inside the live window, a sector-capped twin, and a hold-out window that was not used to tune thresholds. None of that is in this file.

2. Strategy Rules

Rules are unambiguous and applied with no look-ahead. Compliance is a hard constraint: a name that fails on a rebalance date is not held.

2.1 Pipeline at each month-end

1. **Start** from the sector-eligible S&P 500 list (439 names).
2. **Filter (AAOIFI).** Drop names that fail debt / 24-month market cap < 30%, or the cash and receivables screens, using filings known as of that date (`filed_date`, or report date plus 90 days).
3. **Score.** Compute ROIC and reinvestment rate from annual income, balance sheet, and cash-flow statements.
4. **Select.** Require ROIC > 15%, positive NOPAT, positive invested capital, and reinvestment rate > 0. Keep the top half of that Halal list by reinvestment rate (`KEEP_QUANTILE` = 0.50).
5. **Weight.** Cap-weight by market cap. No equal-weight, no inverse-vol, no optimizer.
6. **Rebalance.** Repeat at month-end (`REBALANCE_FREQ` = `ME`).

If fewer than 20 names survive (`MIN_HOLDINGS`), that month is skipped and the book stays in cash (treated as missing, not as a 0% return). Performance statistics start on the first day the strategy actually holds stocks, not on the 2019 start date.

2.2 Entry, exit, and risk limits

Rule	Specification
Entry	Halal + ROIC > 15% + positive reinvestment + top half by reinvestment
Exit	Sold at the next month-end if the name fails AAOIFI, loses the ROIC/reinvestment test, or drops out of the top half
Intra-month stop-loss	None
Position cap	None beyond cap-weight
Cash when thin	Skip the month if holdings < 20
Robustness twin	Same screen, then keep only SaaS / Healthcare / MedTech

There is no stop-loss and no single-name cap. At the last rebalance (31 Dec 2024) Nvidia was 20.4%, Microsoft 19.0%, and the two Alphabet share classes together were about 28.7%. That is a concentrated mega-cap book by design.

2.3 Holdings through time

Median live book size is 51 names (min 5 early build, max 64). The funnel at a typical month is: about 418 names with fundamentals, 100 Halal with ROIC > 15% and positive reinvestment, 50 actually bought. About half of the high-ROIC pool sits in the tagged SaaS / Healthcare / MedTech buckets, but cap-weighting still leaves most weight in "Other" mega-caps.

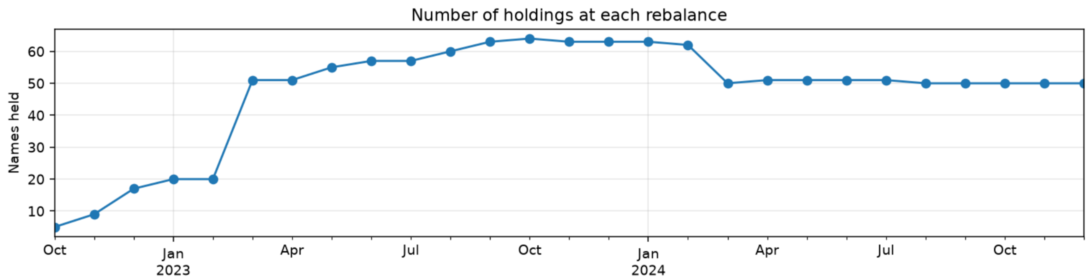

Figure 1: Number of names held at each month-end. The book ramps from a thin early window to a stable 50-name sleeve through 2024.

2.4 Names that enter, stay, or leave

Sharia status, ROIC, and reinvestment rank are re-checked every month. A held name that fails is exited at that rebalance—not left in the book until a human review.

On 31 Dec 2024 the watchlist split as follows:

- **Held:** NVDA, MSFT, GOOGL, ADBE, LLY. All Halal-ok, all in the top half by reinvestment among the 99-name high-ROIC pool.
- **Halal-ok but not bought:** AAPL (ROIC 57% but reinvestment rank 58 of 99—a cash-rich harvester on this metric).
- **Not bought — ROIC floor:** ISRG, SYK, TMO, INTU, NOW, ORCL, and several other MedTech/SaaS names sat below 15% ROIC on Yahoo filings despite strong reinvestment rates.
- **Not bought — negative reinvestment:** UNH and ABBV were cash harvesters that month (reinvestment rate below zero).

GOOG and GOOGL were both held as separate lines. Cap-weighting two share classes of the same company inflates Alphabet relative to a single-listing index. That is a construction bug to fix in a follow-up, not a feature.

Many capital-light MedTech names failed the 15% ROIC floor in this data. That is a central finding for the paper's original sector focus: the reinvestment engine *worked* as a sort inside the high-ROIC pool, but MedTech did not dominate the book the way the backlog hypothesis suggested.

3. Empirical Performance

All headline numbers use the **live window** only: 31 October 2022 through 30 December 2024 (544 trading days). Days before the first invested date are omitted, not filled with zeros. Yahoo's short annual statement history is why the book does not go live in 2019.

3.1 Headline results

Table 1: Live-window performance, 31 Oct 2022 – 30 Dec 2024. Sharpe and Sortino use a 2% risk-free rate. Max drawdown is peak-to-trough on the growth curve.

Book	CAGR	Vol.	Sharpe	Sortino	Max DD	Calmar
Halal ROIC compounders	39.57%	19.48%	1.71	2.57	−15.33%	2.58
Compounders (SaaS / HC / MedTech)	27.61%	17.48%	1.37	1.97	−15.07%	1.83
S&P 500 (SPY)	22.94%	13.92%	1.41	2.13	−9.97%	2.30
Halal (SPUS)	27.35%	16.20%	1.45	2.24	−11.46%	2.39

The compounder book beat both required benchmarks on CAGR, Sharpe, Sortino, and Calmar. It took a deeper drawdown than SPY (−15.3% vs −10.0%) and SPUS (−11.5%). The capital-light robustness sleeve matched SPUS on CAGR (27.6% vs 27.4%) with similar drawdown—it did **not** beat the broad compounder book.

These CAGRs are annualized figures from a mega-cap bull market. They are not a forecast of 40% a year.

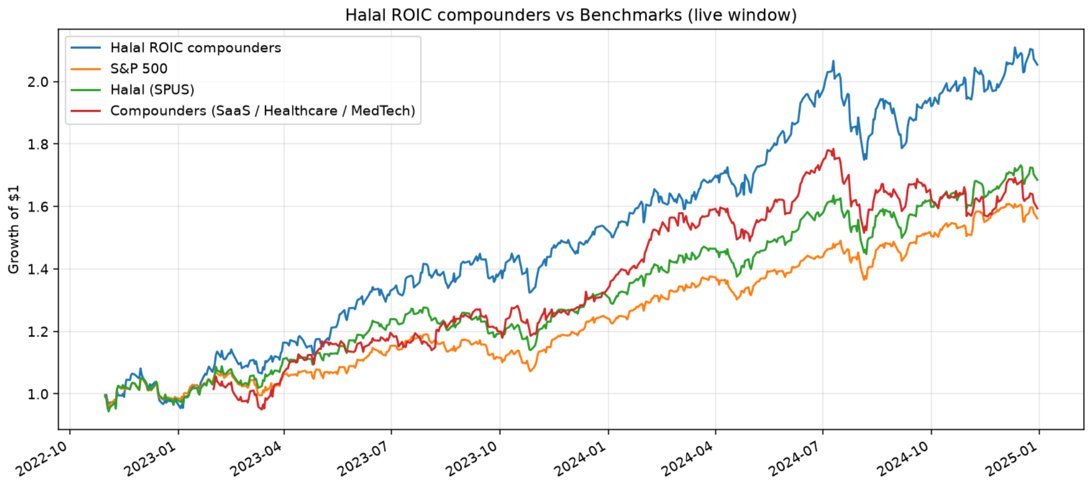

Figure 2: Growth of $1 in the live window. The main compounder book leads; the capital-light sleeve tracks closer to SPUS.

Table 2: Calendar-year total returns inside the live window.

Year	Halal ROIC compounders	S&P 500	Halal (SPUS)
2022	−2.28%	−1.24%	−1.64%
2023	56.67%	26.18%	34.20%
2024	34.16%	25.34%	27.67%

3.2 Calendar years

The compounder book led strongly in 2023 and 2024. 2022 is only a partial year (live from late October) and all books were slightly negative in that stub.

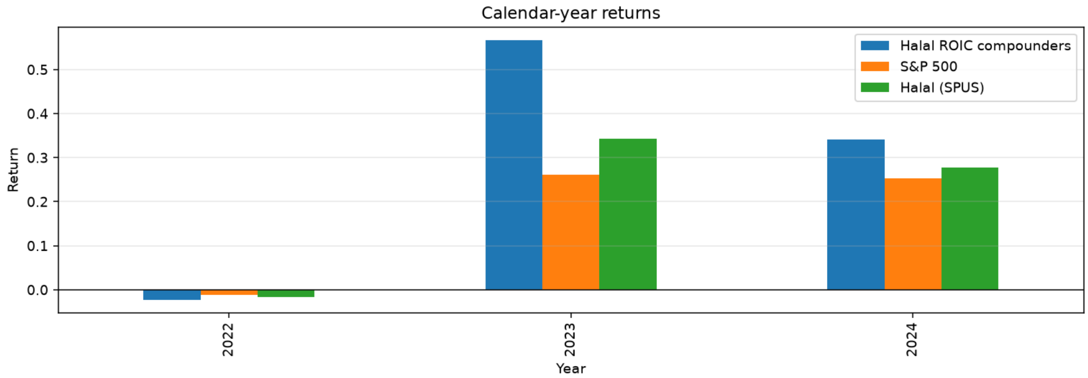

Figure 3: Calendar-year returns. The compounder book is the tallest bar in 2023 and 2024.

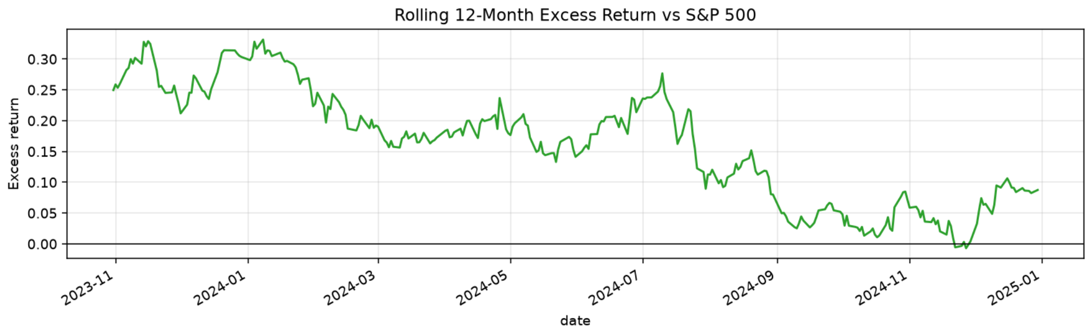

Figure 4: Rolling 12-month excess return versus SPY. Excess stayed positive through most of the live window.

3.3 Drawdowns

The compounder book and SPUS move together in late 2023 and 2024. SPY's drawdowns are shallower. That is consistent with a Halal mega-cap platform book, not with a defensive quality sleeve.

3.4 Does higher ROIC actually compound?

Two cuts inside the Halal universe test the factor logic.

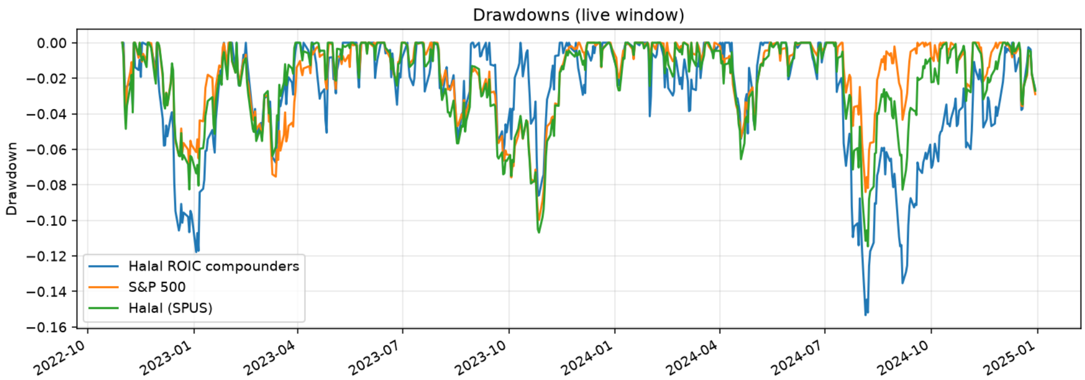

Figure 5: Drawdowns in the live window. The compounder book dips with Halal/tech names and takes a deeper trough than SPY in mid/late 2024.

ROIC quintiles. Cap-weight five buckets by ROIC. Q1 is the highest ROIC; Q5 is the lowest.

Table 3: Cap-weighted ROIC quintiles inside the Halal universe, live window. Q1 = highest ROIC.

Bucket	CAGR	Vol.	Sharpe	Sortino	Max DD	Calmar
Q1	27.09%	18.92%	1.26	1.81	−13.98%	1.94
Q2	33.81%	16.99%	1.68	2.44	−12.61%	2.68
Q3	16.85%	14.17%	1.03	1.55	−14.86%	1.13
Q4	26.23%	18.27%	1.26	1.86	−16.25%	1.61
Q5	18.90%	17.40%	0.97	1.46	−12.89%	1.47

The pattern is **not** monotonic: Q2 beat Q1, and Q4 beat Q5. Q1−Q5 annualized spread was 6.9%, with Q1 beating Q5 on 52.4% of days. ROIC alone is a noisy sort in this window.

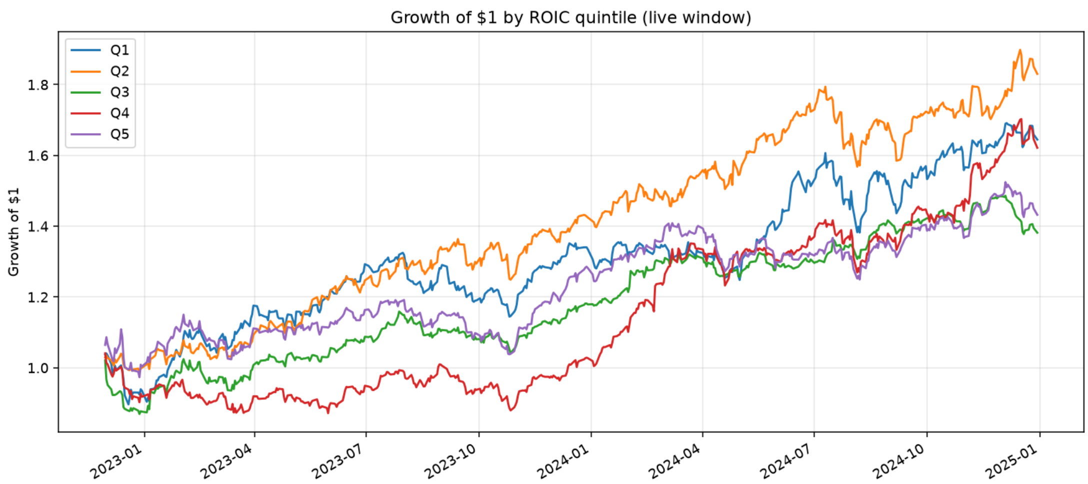

Figure 6: Growth of $1 by ROIC quintile. Q2 leads; Q3 lags.

Reinvestment split among high-ROIC names. Hold ROIC > 15% fixed and split the pool at the median reinvestment rate. This is the compounding hypothesis the live rule implements.

High reinvestment beat low reinvestment by 14.5 points of CAGR—stronger evidence than the quintile sort. The live book targets the high-reinvestment half and lands near this sleeve by construction.

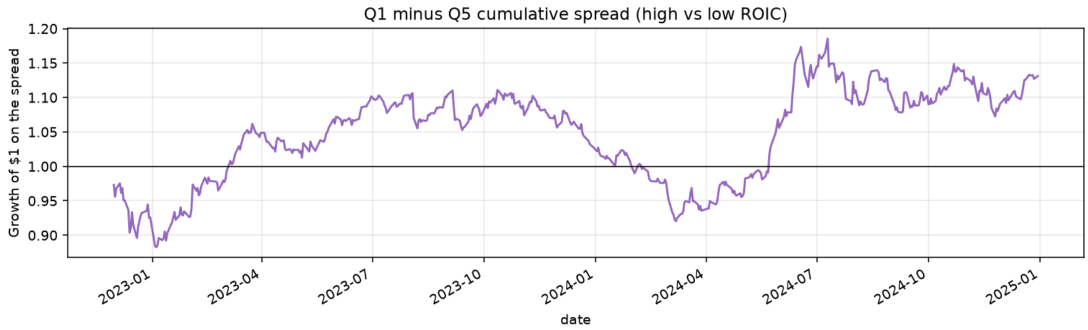

Figure 7: Cumulative Q1 minus Q5 spread. The line ends above 1 but is noisy—not a clean stair-step quality premium.

Table 4: Reinvestment split inside the Halal ROIC$>15\%$ pool, live window, cap-weighted.

Sleeve	CAGR	Vol.	Sharpe	Sortino	Max DD	Calmar
High ROIC + high reinvestment	43.18%	19.34%	1.85	2.74	−15.33%	2.82
High ROIC + low reinvestment	28.71%	14.23%	1.70	2.88	−11.45%	2.51

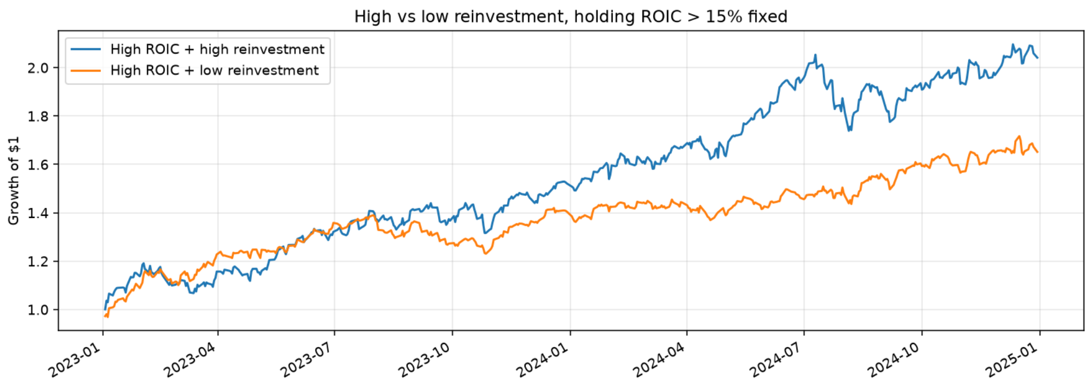

Figure 8: High vs low reinvestment, holding ROIC $> 15\%$ fixed. The high-reinvestment sleeve pulls ahead through 2023–2024.

4. Factor Attribution

Section 3 shows the book won. This section asks whether that win is stock picking or a hidden sector bet—especially the SaaS / Healthcare / MedTech tilt the backlog named.

4.1 Alpha, beta, tracking error

Daily excess returns of the compounder book are regressed on each benchmark (CAPM-style, 2% risk-free rate). Alpha is annualized.

Table 5: Halal ROIC compounders versus each benchmark, live window.

Benchmark	Alpha (ann.)	Beta	R^2	Tracking error	Info. ratio
S&P 500	10.00%	1.18	0.72	10.68%	1.28
Halal (SPUS)	8.06%	1.07	0.80	8.89%	1.10

Versus SPY the book is more aggressive ($\beta = 1.18$) with 10.0% leftover annualized alpha. Versus SPUS it is almost a parallel book: $\beta = 1.07$, $R^2 = 0.80$, still 8.1% alpha. In plain language: **this is an SPUS-like Halal large-cap book with extra mega-cap platform exposure**, not an independent style that wiggles on its own.

The information ratios (1.28 vs SPY, 1.10 vs SPUS) look respectable because the sample includes two strong years. They will shrink if a full down year like 2022 is added.

4.2 Focus-sector and GICS exposure delta

Average month-end weights of the cap-weighted compounder book are compared with an equal-weight basket of the Halal S&P universe.

Table 6: Average focus-bucket weights. Delta = compounder book minus equal-weight Halal universe.

Bucket	Compounder book	Halal universe (EW)	Delta
Other	73.7%	77.7%	−4.0%
SaaS	16.6%	9.1%	+7.5%
Healthcare	9.4%	5.7%	+3.7%
MedTech	2.0%	7.5%	−5.5%

SaaS and Healthcare are overweights, but MedTech is an **underweight**. Cap-weighting Nvidia, Microsoft, Alphabet, and Meta leaves “Other” at 74% despite many tagged names in the high-ROIC pool.

Technology plus communication services average **82.4%** of the book—even more concentrated than Paper 01’s FCF-quality book. Utilities, real estate, and financial services are empty.

4.3 What we can and cannot isolate

The leftover alpha versus SPUS is still positive after a $\beta \approx 1$ fit, so the result is not *only* “we owned SPUS.” But SPUS itself is already Halal and tech-tilted in this window, and our book is

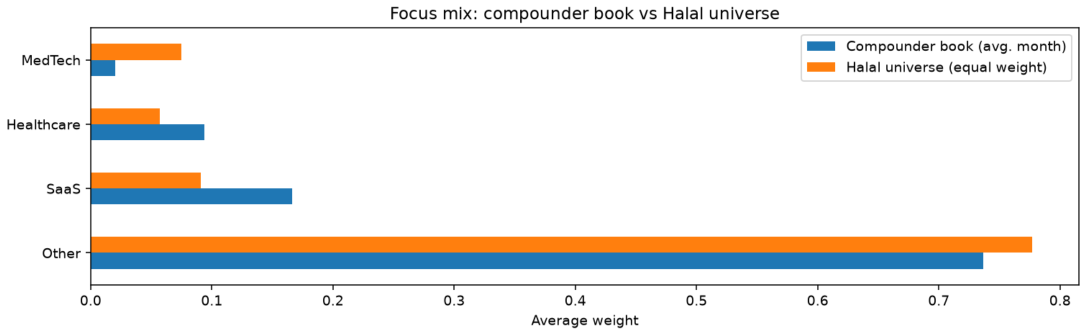

Figure 9: Focus mix: compounder book vs equal-weight Halal universe. MedTech is lighter than the universe, not heavier.

Table 7: Average GICS sector weights. Delta = compounder book minus equal-weight Halal universe.

Sector	Compounder book	Halal universe (EW)	Delta
Technology	44.6%	19.1%	+25.5%
Communication services	37.8%	5.5%	+32.4%
Healthcare	9.7%	13.4%	−3.8%
Consumer cyclical	6.8%	11.6%	−4.8%
Consumer defensive	5.5%	6.4%	−0.9%
Energy	2.7%	4.8%	−2.1%
Industrials	1.9%	14.4%	−12.4%
Basic materials	1.1%	4.6%	−3.4%
Utilities	0.0%	7.1%	−7.1%
Real estate	0.0%	6.8%	−6.8%
Financial services	0.0%	6.2%	−6.2%

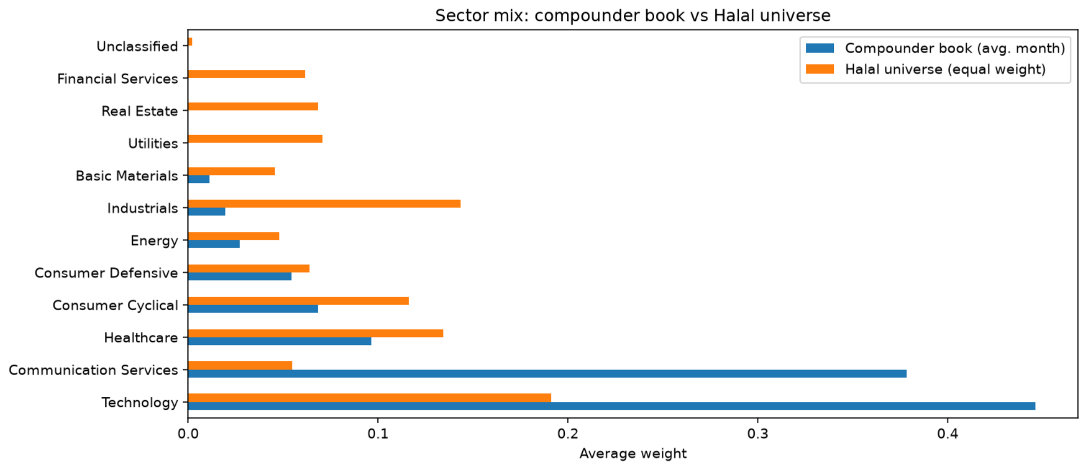

Figure 10: GICS sector mix: compounder book vs equal-weight Halal universe. The book is a platform/tech overweight, not a diversified capital-light healthcare book.

more concentrated in the same mega-cap platform names.

The reinvestment split supports the *engine* part of the hypothesis: among high-ROIC Halal names, ploughing capital back in paid in 2022–2024. The *sector* part is weaker: MedTech did not dominate, and the SaaS/HC/MedTech-only sleeve did not beat the main book.

Without a sector-neutral twin we cannot say how much of the 8% SPUS alpha is reinvestment sorting versus "even more Nvidia/Microsoft/Alphabet/Meta than SPUS."

The honest attribution: **reinvestment sorting inside a Halal high-ROIC pool produced a mega-cap platform book, and that book paid in 2022–2024.**

5. Limitations & Friction

5.1 Turnover, slippage, and net CAGR

Turnover is one-way: half the ℓ_1 weight change between consecutive month-end books. Trading-cost drag assumes 10 basis points per one-way turn.

Table 8: Implementation friction on the live compounder book.

Item	Value
Rebalances with holdings	27
Average names held	47.6
Annualized one-way turnover	200.2%
Assumed cost per one-way trade	10 bp
Estimated annual trading-cost drag	0.20%
CAGR gross of trading costs	39.57%
CAGR net of trading costs	39.24%

Annualized one-way turnover of 200% looks high because the book ramps from a thin early window and rebalances monthly across 50 names. Cost drag at 10 bp is 20 bp a year—small next to the headline CAGR, but real slippage on smaller reinvestment picks could be wider. We did not model market impact, borrow, or taxes.

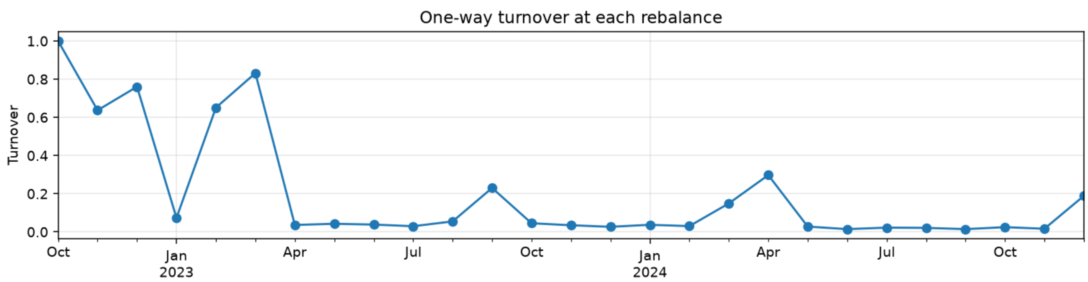

Figure 11: One-way turnover at each rebalance. Spikes appear when the book expands; steady-state months are lower but not negligible.

5.2 Purification drag

Impure income that may require purification is reported, not ignored. Using the library's dividend-purification helper on names that appeared in the book: **213 dividend events while held**, annualized purification drag $\approx$ **0.00**%. That is negligible next to 39.6% CAGR. Missing interest-income tags are a gap, not a 0% drag.

5.3 Point-in-time compliance and exits

Screens are applied at each rebalance. Failed names are sold then. There is no intra-month compliance monitor.

Sector exclusions use the *current* S&P list and current activity labels, not the 2019 membership file. That is survivorship and look-ahead in the universe, even though filings themselves use `filed_date`.

5.4 Data and construction issues

- **Short statement history.** Yahoo annual statements typically store a few years. First metrics date is 30 Nov 2021; first invested date is 31 Oct 2022. The 2019–2021 request is mostly unused.
- **ROIC quality.** NOPAT, invested capital, CapEx, D&A, ΔNWC, and R&D come from yfinance with a 90-day lag. Restatements and tag mapping are not audited here. Names with ROIC above 500% are dropped as denominator errors.
- **Share-class double count.** GOOG and GOOGL are both weighted. Alphabet is overweight versus a single-listing index.
- **No position cap.** Three names can be half the book. A 5% or 8% cap would change both risk and 2023–2024 return.
- **MedTech ROIC floor.** Many med-tech names failed the 15% ROIC threshold on Yahoo data despite high reinvestment. Sector persistence cannot be read from weights alone.
- **15% ROIC threshold.** Fixed before the full run; not optimized on a hold-out, but the live window is short.

5.5 What this note does not show

There is no full calendar year 2022, no 2008, no rate-hike recession, and no hold-out after the rule freeze. ROIC quintiles are not monotonic. Sector delta explains a large part of the path. Beating SPUS in a mega-cap boom is interesting; it is not evidence that Halal ROIC compounders persist across cycles in capital-light sectors—the original backlog question.

5.6 Decision

Follow-up, do not kill, do not promote to a live book.

Keep the idea: Halal + high ROIC + high reinvestment + cap-weight is a coherent product shape, and the reinvestment split supports the engine logic more than ROIC quintiles alone. Before a second paper, freeze the rule and add (i) SEC income/cash-flow tags with a longer history, (ii) point-in-time S&P members, (iii) a single listing per issuer, (iv) a sector-capped or MedTech-inclusive twin with a ROIC definition tuned for R&D-heavy filers, and (v) at least one full down year. If the sector-capped twin still beats SPUS, the compounding claim gets teeth. If it does not, this note's 40% is mostly the 2023–2024 platform tape—similar to Paper 01.

Reproduction. Notebook `Research/papers/02-roic-engine/code.ipynb` with `FAST_MODE=False`, 2019-01-01 to 2024-12-31. Figures in `figures/`.

References

[1] AAOIFI, "Financial Paper (Shares and Bonds)," Shari'ah Standard No. 21, Accounting and Auditing Organization for Islamic Financial Institutions.

[2] Monterey Finance, "Halal Quant Research Lab," `Research/README.md`, Phase 1 standardized white-paper structure.